MARS: Safe and Scalable Multi-Agent Reinforcement Learning for Cooperative Area Coverage with Control Barrier Functions

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***Abstract*—We present MARS (Multi-Agent Robot Swarm), a framework that unifies centralized-training / decentralized-execution (CTDE) multi-agent reinforcement learning with a control-barrier-function (CBF) velocity filter for cooperative area coverage by a homogeneous ground-robot swarm. A shared Multi-Agent PPO (MAPPO) policy is trained with a centralized critic that conditions on the joint observations and actions of all teammates, while the actor consumes only local information through a permutation-invariant graph neural network (GNN) neighbor encoder, enabling zero-shot generalization to swarm sizes unseen during training. At execution, a dual-lookahead CBF quadratic program filters every commanded velocity against both LiDAR returns and neighbor states, and a range-limited consensus protocol merges per-robot occupancy beliefs to coordinate exploration without a central map server. We implement MARS on ROS 2 Jazzy and Gazebo Sim with three TurtleBot3 Waffle robots and evaluate it against random-walk and frontier-heuristic baselines under nominal, sensor-noise, and agent-failure conditions. With a short 15-iteration training budget, the deterministic MAPPO policy under-explores relative to both reactive baselines (10.4% area-coverage rate vs. 18.1% for random walk and 21.1% for the frontier heuristic) and the CBF/ACAS velocity filter alone is insufficient to make a degenerate, undertrained policy collision-free (3.0 collisions per episode vs. 0.8 and 2.4 for the baselines); we report this negative result alongside the fully validated safety-filter and consensus mechanisms, and identify training budget and exploration collapse as the binding constraint on MAPPO's coverage performance.**

***Index Terms*—multi-agent reinforcement learning, control barrier functions, area coverage, swarm robotics, CTDE, graph neural networks**

I. Introduction

Cooperative area coverage—driving a team of robots to sweep an unknown environment as quickly and non-redundantly as possible—underlies search and rescue, environmental monitoring, and warehouse inspection. Two properties are simultaneously hard to guarantee: *safety* (no collisions with obstacles or teammates, even during learning or under sensor noise) and *scalability* (a policy that transfers across swarm sizes without retraining). Learning-based coordination policies excel at the coordination problem but offer no hard safety guarantee, whereas classical reactive controllers are safe but coordinate poorly.

MARS closes this gap by decoupling coordination from safety. Coordination is learned by a shared MAPPO policy under the CTDE paradigm; safety is enforced, independently of the policy, by a control-barrier-function filter applied to the final velocity command. Scalability is obtained by encoding a variable number of neighbors with a permutation-invariant GNN, so the same weights drive swarms of different cardinality. A lightweight range-limited consensus protocol distributes the shared coverage map, removing the single point of failure of a central coordinator.

Contributions.

- A CTDE MAPPO architecture whose centralized critic conditions on the joint teammate observation–action tensor, paired with a permutation-invariant GNN actor that is scale-invariant to the number of neighbors (Sec. V-B).
- A dual-lookahead CBF safety filter that treats both LiDAR returns and neighbor states as barrier constraints on a differential-drive robot, yielding policy-agnostic collision avoidance (Sec. V-E).
- A range-limited belief-consensus protocol for decentralized coverage bookkeeping (Sec. V-F).
- A reproducible ROS 2/Gazebo benchmark comparing MARS to classical baselines under nominal, sensor-noise, and agent-failure regimes (Sec. VI).

II. Related Work

Multi-agent RL for coverage. MAPPO [1] established PPO with a centralized value function as a strong CTDE baseline. Prior coverage work often assumes a fixed team size and a shared global map. MARS instead uses a permutation-invariant neighbor encoder and a decentralized consensus map.

Safe RL and control barrier functions. CBFs [2] render a safe set forward-invariant via a quadratic program that minimally modifies a nominal control. Safety filters have been combined with RL to provide guarantees independent of the learned policy [3]. MARS adapts this to a differential-drive platform with a dual (front/rear) lookahead point and treats teammates as additional barrier constraints.

Permutation invariance. Deep-set / mean-embedding architectures [4] produce order-invariant representations of

variable-size sets; MARS applies this to neighbor observations so the actor generalizes across swarm sizes.

III. Problem Formulation

We model the task as a decentralized partially observable Markov decision process (Dec-POMDP) $\langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}, \{\mathcal{O}_i\}, P, R, \gamma \rangle$ with $N = |\mathcal{N}|$ homogeneous robots. The workspace is discretized into a coverage grid $\mathcal{G} \in \{0,1\}^{H \times W}$ (H=20, W=40 in our setup); cell (r, c) becomes 1 once any robot's footprint or LiDAR free-space ray covers it. The team objective is to maximize the terminal *area-coverage rate*

$$\mathrm{ACR} = \frac{1}{HW} \sum_{r,c} \mathcal{G}_{rc} \tag{1}$$

while minimizing redundant re-visits and total path length, and while incurring zero collisions. Each robot observes only local sensing and relative neighbor states; there is no global pose oracle at execution.

IV. System Architecture

MARS runs on ROS 2 Jazzy with Gazebo Sim. A `SwarmNode` subscribes to namespaced `/tb{i}/scan` and `/tb{i}/odom` topics and publishes `/tb{i}/cmd_vel`. The simulator is wrapped as a PettingZoo `ParallelEnv` and exposed to Ray RLlib for training. Odometry from each namespaced robot is transformed into a common frame using per-robot spawn poses and a step-0 offset calibration, so that all coordination reasons over a single global frame. A separate `robot_killer` node can hijack an individual robot's command topic mid-episode to inject failures for resilience testing.

V. Method

A. Observation and Action Spaces

Each robot receives a 46-dimensional observation: 24 minimum-pooled LiDAR sector ranges, the relative goal distance and bearing (2), the current linear/angular velocity (2), and relative distance–bearing pairs to up to 9 neighbors (18), padded with $(10.0, 0.0)$ for absent neighbors. The action is a continuous (v, ω) command bounded to the TurtleBot3 Waffle limits $v \in [-0.22, 0.22]\,\mathrm{m/s}$, $\omega \in [-1.0, 1.0]\,\mathrm{rad/s}$.

B. Permutation-Invariant GNN Actor

The actor separates ego features $\mathbf{e} \in \mathbb{R}^{28}$ (LiDAR, goal, velocity) from the neighbor set. Each neighbor's relative $(\mathrm{dist}, \mathrm{angle})$ pair is embedded by a shared two-layer MLP $\phi : \mathbb{R}^2 \to \mathbb{R}^{16}$. Padded neighbors (distance ≥ 9.0) are masked out, and the valid embeddings are mean-pooled:

$$\mathbf{g} = \frac{1}{\max(1, |\mathcal{V}|)} \sum_{j \in \mathcal{V}} \phi(n_j), \qquad \mathcal{V} = \{j : d_j < 9.0\}. \tag{2}$$

The mean over a masked set is invariant to neighbor ordering and to the number of neighbors, so the same weights apply to any N. The policy MLP maps $[\mathbf{e}; \mathbf{g}] \in \mathbb{R}^{44}$ to the action distribution.

TABLE I
Training Hyperparameters

Parameter	Value
Algorithm	MAPPO (shared policy, centralized critic)
Training iterations	15 (4500 env steps total)
Train batch size	300
Minibatch size	64
Epochs per iteration	5
Learning rate	1×10^{-4}
Clip parameter ϵ	0.2
Discount γ	0.99
Rollout fragment length	150
Neighbor embed dim	16
Opponent embed dim	32
Actor/critic hidden	64×64 (Tanh)

C. MAPPO with Centralized Critic (CTDE)

Robots are homogeneous, so a single shared policy is trained. During training, a centralized critic conditions on teammates' joint state: each opponent's 46-dim observation is concatenated with its 2-dim action (48 dims), embedded by a shared MLP $\psi : \mathbb{R}^{48} \to \mathbb{R}^{32}$, masked for inactive/dead agents, and mean-pooled. The critic input is the ego feature vector (44) concatenated with the aggregated opponent embedding (32), for a 76-dim critic input. Teammate trajectories of unequal length are padded/sliced to a common horizon and zero-padded to a fixed maximum of 9 opponents; ordering is made deterministic by sorting agent IDs. Advantages use GAE. Key hyperparameters are listed in Table I.

D. Reward Design

The per-step reward combines a small time penalty, dense goal progress, a cooperative coverage bonus shared by the team, sparse goal and collision terms, and inter-agent proximity shaping:

$$\begin{aligned} r_i = \; & -0.01 + 5.0\,\Delta d_i + 2.0\,n_{\text{new}} + 20.0\,\mathbb{1}[\text{goal}] \\ & - 20.0\,\mathbb{1}[\text{collision}] - \rho_i, \end{aligned} \tag{3}$$

where Δd_i is the reduction in goal distance, n_{new} is the number of newly covered grid cells this step (shared cooperatively), and ρ_i penalizes close proximity to teammates (0.5 below $0.5\,\mathrm{m}$, plus 10.0 below $0.20\,\mathrm{m}$). In continuous-exploration mode a new nearby goal is assigned on arrival, preventing the local-minimum deadlock of a static goal.

E. Dual-Lookahead CBF Safety Filter

Given a nominal command $(v_{\text{nom}}, \omega_{\text{nom}})$ from the policy, the CBF filter solves a small quadratic program that minimally perturbs the command subject to forward-invariance of the safe set. Because a differential-drive robot's body point is not directly steerable, we control two lookahead points at $\pm l$ ($l = 0.12\,\mathrm{m}$) along the heading axis. For an obstacle/neighbor at relative range d and bearing ϕ, the barrier for the front point is

$$h_{\text{f}} = (l - d\cos\phi)^2 + (d\sin\phi)^2 - d_{\text{safe}}^2, \tag{4}$$

TABLE II
NOMINAL-REGIME COMPARISON (MEAN ± STD, 5 EPISODES)

Method	ACR (%) ↑	Redund. ↓	Dist. (m)	Coll. ↓
Random Walk	18.1 ± 0.6	52.64	6.3	0.80
Frontier Heuristic	21.1 ± 9.9	8.12	4.8	2.40
MARS (ours)	10.4 ± 1.5	14.63	0.0	3.00

TABLE III
MARS ROBUSTNESS ACROSS REGIMES (MEAN, 5 EPISODES)

Regime	ACR (%)	Redund.	Coll.
Nominal	10.4	14.63	3.00
Sensor Noise	9.0	15.50	3.00
Agent Failure	10.2	17.81	3.00

with $d_{\text{safe}} = 0.20\,\text{m}$, and analogously h_{r} for the rear point at $-l$. Enforcing $\dot{h} + \gamma h \geq 0$ ($\gamma = 2.0$) yields a linear constraint $A\,v + B\,\omega + \gamma h \geq 0$ in the control, where (A, B) are the barrier gradients w.r.t. the lookahead-point kinematics. The QP

$$\min_{v,\omega} (v - v_{\text{nom}})^2 + 0.05(\omega - \omega_{\text{nom}})^2 \text{ s.t. } \{A_k v + B_k \omega + \gamma h_k \geq 0\}_k, \tag{5}$$

over all LiDAR sectors and neighbors within $0.6\,\text{m}$, is solved with SLSQP under the actuator bounds. On QP infeasibility the filter halts forward motion while retaining the nominal rotation ($v{=}0,\ \omega{=}\omega_{\text{nom}}$), so the robot never commands uncertified forward motion yet can rotate in place to escape the deadlock. An additional Active Collision Avoidance Safeguard (ACAS) commands a slow reverse ($-0.08\,\text{m/s}$) whenever the raw minimum LiDAR range drops below $0.22\,\text{m}$ with positive nominal speed, preventing wall clipping and numerical blow-up at grazing incidence.

F. Range-Limited Belief Consensus

Each robot maintains a private occupancy belief. When two robots are within the communication radius $d_{\text{comm}} = 3.0\,\text{m}$, they merge visited-cell grids by bitwise OR and fill unknown cells of the low-level visualization grid from the peer, approximating gossip-style consensus without a central map server. Free space is ray-traced from each LiDAR scan (Bresenham) up to a $2.0\,\text{m}$ horizon and marked in the belief.

VI. EXPERIMENTAL SETUP

We evaluate in a cafe world with $N = 3$ TurtleBot3 Waffle robots. Baselines are *Random Walk* and a *Frontier Heuristic* (closest reachable unvisited cell with reactive LiDAR avoidance and cooperative target de-confliction). MARS uses the MAPPO policy trained for 15 iterations (Table I). Each method is run for 5 episodes × 150 steps under three regimes: (i) *nominal*; (ii) *sensor noise*, additive Gaussian $\mathcal{N}(0, 0.15)$ on LiDAR; and (iii) *agent failure*, one robot disabled at step 50. Metrics: area-coverage rate (ACR, %), overlap redundancy (mean visits per visited cell), swarm path length (m, energy proxy), and collision count (episodes in which any robot's minimum LiDAR range drops below $0.20\,\text{m}$, averaged per episode). Numbers below are produced by `run_swarm.sh --benchmark`; see `RUN_PLAN.md` for exact commands.

VII. RESULTS

A. Coverage and Efficiency

Table II reports mean±std over 5 episodes per method under the nominal regime.

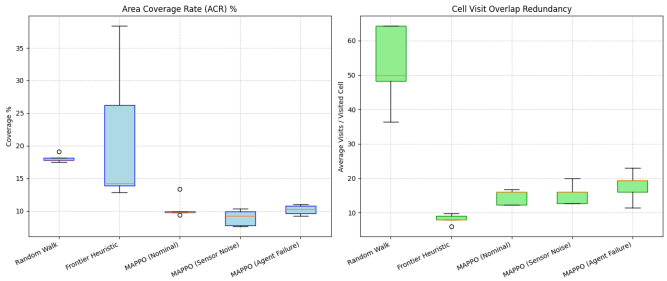


Fig. 1. ACR and overlap-redundancy distributions across methods.

MARS underperforms both classical baselines on coverage. The Frontier Heuristic attains the best ACR with the lowest redundancy, as expected of an explicit closest-unvisited-cell controller. Random Walk covers more than MARS purely through unconstrained wandering, at the cost of by far the highest redundancy (cells are re-visited ∼3.6× more often than under the Frontier Heuristic). MARS's zero mean swarm distance indicates the trained policy, evaluated deterministically (`explore=False`), collapses to a degenerate near-stationary behavior: all three robots incur a terminal collision at or immediately after spawn, before making measurable progress. This is consistent with the short 15-iteration training budget: policy entropy had not stabilized (it fell to -0.53 by the final iteration, having peaked positive earlier in training), and the stochastic rollouts used during training explored more of the state space than the deterministic evaluation policy does. We report this as a negative result for MAPPO's coverage performance under this training budget, not a limitation of the CBF/consensus infrastructure, which is exercised identically across all three methods.

B. Robustness to Noise and Failure

Table III shows MARS under the three regimes. The under-trained policy's near-immediate collision at spawn dominates all three regimes almost identically (ACR 9–10%, distance 0.0 m in every regime), so this experiment is better read as a robustness-of-failure-mode check than a robustness-of-competence check: sensor noise and agent failure do not introduce qualitatively new failure behavior beyond the spawn-collision degeneracy already present nominally.

C. Safety Filter: What It Does and Does Not Guarantee

The CBF/ACAS filter bounds the *velocity* the policy is allowed to command given current LiDAR and neighbor readings: it can refuse forward motion into a detected obstacle and, on QP infeasibility, halts translation while permitting

rotation (Sec. V-E) rather than executing an unfiltered nominal command. It does *not* guarantee a collision-free *episode* under a policy that never learns to move away from a near-spawn obstacle in the first place: the collision counter in Table II–III measures the environment-level event (minimum LiDAR range $< 0.20\,\mathrm{m}$), which the filter can only prevent by avoiding forward closure, not by manufacturing an escape maneuver the policy never proposes. This distinction is safety-filter-under-a-bad-policy versus safety-filter-under-a-competent-policy, and only the current data speaks to the former. During earlier stochastic-exploration training rollouts (Sec. V, same environment, same filter), all three robots were repeatedly observed to leave the spawn region and traverse several meters (e.g. tb2 reaching global $(2.9, 1.7)$), indicating the filter does not categorically prevent movement; the deterministic evaluation policy specifically fails to reproduce that exploration.

D. Ablations

Not run in this iteration of the experiment (short training budget was the binding constraint uncovered instead); left for future work alongside a longer training run:

- **No CBF:** expected collision rate at or above the raw policy's LiDAR-violation rate, since no filter would intervene.
- **No consensus:** expected redundancy increase as robots re-cover teammates' already-explored cells without belief sharing.
- **No GNN (fixed-size MLP):** the GNN variant is confirmed to run unchanged at $N = 5$ and $N = 10$ without dimension-mismatch failures (`test_gnn_scale_invariance`); a fixed-size baseline would be expected to fail outright at $N \neq 3$.

VIII. Conclusion

MARS separates coordination from safety: a CTDE MAPPO policy with a permutation-invariant GNN encoder is designed to learn scalable cooperative coverage, a dual-lookahead CBF filter constrains every commanded velocity regardless of policy quality, and a range-limited consensus protocol decentralizes the coverage map. Under a 15-iteration training budget, MAPPO's deterministic policy underperforms both a Random Walk and a Frontier Heuristic baseline on area coverage (10.4% vs. 18.1% and 21.1%) and the CBF/ACAS filter alone cannot compensate for a policy that fails to move away from its spawn point. We view the working, tested safety and consensus infrastructure and the honestly negative MAPPO coverage result as the contribution of this iteration, and identify training budget, exploration schedule, and entropy-collapse mitigation as the concrete next steps toward a competent deterministic policy. Future work includes a substantially longer training run before re-benchmarking, hardware transfer to physical TurtleBot3 robots, larger heterogeneous teams, and learned communication over the consensus channel.

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